Project (150 pts)

MATH 241 — Probability, Fall 2026

Background

Students will work in groups of 2-3 to not just answer a probability question, but to understand the whole context behind the question, the technology used to answer the question, and how it can inform our understand of now, the past, and/or the future. You will not necessarily just answer the probability question – you will investigate how we can even answer it.

By the end of this project, you will be able to do the following:

- Connect concepts, formulas, considerations, and methods from class to answer a real-world probability question.
- Discover historical context with the real-world impacts of a probability question.
- Use data, simulations, and/or methods to answer the probability question.
- Analyze and report on findings using clear and concise language through a report and demonstration.

Your group will accomplish all of the aforementioned tasks through the following parts:

1. **Proposal**: 5 pts, due Friday, 10/30 at 11:59pm
    - Consists of forming groups and writing a project proposal.
2. **Progress Checkpoint**: 15 pts, due Friday, 11/20 at 11:59pm
    - A (very) rough draft of your written report
    - A 1 page summary of what has been accomplished and what will need to be done over the next few weeks
3. **Demonstrations**: 40 pts, given either during class time or away from class on Tuesday, 12/8
    - A *8-10 minute demonstration* in an out-of-class office visit discussing the main points of your project.

4. **Final Portfolio**: 90 pts, due Wednesday, 12/9 at 11:59pm
    - A *4-6 page written report* containing all requirements in APA format (e.g. 12-point font, Times New Roman, double-spaced).
    - A script/Quarto document of R code showing all models and analysis done.

Breakdown of Project

Your group will be tasked with exploring a probability question of your choosing, rooted in some discipline, hobby, or interest. This question should be complex enough to where an answer should not be determined quickly.

> Example: A simple question might be, "what is the likelihood of getting at least 10 heads during 20 coin flips?" Perhaps, we might be curious about the true equal chances of flipping a coin, in which we might question, "how random is a coin flip?". Or, we might question how the use of coin flips may influence certain events, such as events in overtime of a game of American football and political caucuses.

The premise is that the question should, at the very least, make you ponder about the result longer than you should, and it will take a combination of research, simulations, and experimenting to at least attempt to answer the question.

To that end, here are some guiding questions as you are researching your question:

- What has been researched about the topic? Have any experiments been done to answer the question? If so, how did they come to that conclusion? (i.e. history of question)
- Which factors/variables, situations, and other items may influence the way we approach the question? What are the impacts of answering said question? (i.e. real-world context)
- How can you all as a group answer the question? Would you need to perform some simulations and/or work through some arithmetic to get to solutions? How may others be able to come to the same conclusion? (i.e. replication and experimenting)
- What is the solution (if there is one)? How can this be applied to certain situations and/or the future? (i.e. solution and impacts)

While not an exhaustive list of questions to ponder, these will allow you to research your question throughout the rest of the semester, culminating in a demonstration and a final report. For both items, your group must include a(n):

1. Introduction: Explain the premise of the question, what the question is, and initial thoughts
2. Background: Talk about the historical background of the question, its impacts before and now, and (famous) instances of when this question might have been useful to answer
3. Experiment: Hypothesize what the answer could be, discuss a method to solving said question, and conduct that method (either as a simulation or deterministic approach
4. Analysis: Discuss what results you gathered and how those questions can be answered in various manners
5. Conclusion: Summarize what your group has found and how the solution can be useful

For the first two parts, you will look up sources to inform your understanding of the question itself. **You must cite these sources, even if they are synthesized from artificial intelligence.** Particularly when using any of the LLM models, **verify** its legitimacy by asking for and reading through the sources.

For experimenting, this should be rooted in concepts we've discussed as a class throughout the semester. You might need to use a Bayesian framework (e.g. conditional probability, priors), distributions (e.g. exponential, normal), Monte Carlo simulations (e.g. to get a metric or parameter estimate), and other items. Your group will showcase a way to answer the question with these concepts, and this might be done deterministically or through simulation. Your thinking or code must be written-up or coded as part of the project.

For the last two parts, you will be discussing the results and how we can make sense of that value (as well as teaching/informing those reading your work how to interpret the results and apply them to various situations. Ultimately, this should be based on **your own interpretations** of the results and what they might mean to others.

Note about AI

I almost guarantee that for whatever question your group decides on, some LLM will have an answer to it, showcasing how it answered it. I do not want a LLM's interpretation of your probability question, I want **your** interpretation of it.

I am okay with the use of LLMs in various aspects, such as identifying sources or events, initially determining models, and code debugging. However, you will need to be familiar with and explain the foundations and work associated with finding solutions to

your probability question. Your interpretations of the research and experimentation will be more impactful to me than a generic, cookie-cutter response.

Components

***1. Proposal* (*5 pts; due Friday, 10/16 at 11:59pm on Gradescope*)**

For this component, you will submit a 1 page document (Times New Roman, double-spaced, without APA formatting) containing the following:

- (1 pt) A list of the names of all group members.
  - There must be either 2 or 3 group members. You may not work alone or groups more than 3.
  - You are free to work with students in the other section.
- (2 pts) Introduce the probability question your group wants to explore, as well as any relevant background or information to the question.
- (2 pts) A plan regarding what you plan to investigate, which probability concepts may be important here, and how you might solve the problem at first glance.

Have only one person in your group turn in this plan. Comments to your group's plan will be given through Gradescope once it is submitted. Proposals not submitted by the deadline will only count as a 0.5^d proportion of the max proposal grade, where d is the number of days it is late by.

2. Progress Checkpoint* (*15 pts; due Friday, 11/20 at 11:59pm on Gradescope

This progress checkpoint can be treated as a super rough draft of the final report you will turn in at the end of the semester. Quite simply, this should just be a rough version that is at least one page long consisting of a mixture of paragraphs, figures, and formatting related to your investigation. You will preface this rough draft with a summary of items/tasks your group has already done, and what your group is planning to do over the next few weeks. As such, I will be looking for the following:

- (5 pts) A 1 page summary of the following:
  - What your group has done
  - What your group still needs to do

  - Any difficulties or questions your group has up to this point

- (10 pts) Evidence of progress, consisting of
  - Unfinished figures
  - Some paragraphs relating to one the five sections mentioned above
  - Code that has been completed

3. Demonstration (*40 pts; during 10-15 minute slot on Tuesday, 12/8*)

Your group will sign up for a 15 minute slot on Tuesday, December 7th, with sign-ups happening the week prior. There will be slots both during lecture and outside of lecture. A lecture will not be held on this date.

In the demonstration, your group will have 5-8 minutes to discuss your research question and how your group approached the question, showing the process, results, and conclusions.

The demonstration itself can take on the form of a variety of formats. This could be:

- A showcase of your code and methods, discussing how your group approached the question
- A traditional slideshow, with sections similar to what is seen in the report
- Boardwork, showing the setup of the research question and how you all tried to solve the problem

After the 5-8 minute presentation, the remaining time will just be spent answering questions and clarifying items from your investigation. You will be graded on the following items:

- (10 pts) Delivery: points mentioned clearly and succinctly; done so in an organized manner
- (10 pts) Content: arranged well; content is accurate and contains main points; goes through logical and clear order
- (10 pts) Intuition: content delivered with a personable and strong connection to topic; grasps details of their investigation without any aids
- (10 pts) Participation: each member has some role in presentation

***4. Final Portfolio* (*90 pts; due Wednesday, 12/9 at 11:59pm on Gradescope*)**

The final portfolio will consist of two items: a final report and all code associated with the project. This will all be put in a zipped folder, which you will turn into Gradescope.

The project report should summarize all of the work completed. Formatting-wise, your report should:

- (2.5 pts) Contain 4-6 pages (with no abstract or table of contents required; title-page not included in page count)
- (2.5 pts) Adhere to all requirements in APA format (e.g. 12-point font, Times New Roman, double-spaced) and is organized into multiple, sensible sections (with headers indicating each section)

The report should have the following parts:

- (7.5 pts) **Introduction:** An overview of the research question and initial thoughts
- (15 pts) **Background:** The historical background of the question; impacts before and now; instances in which question is important.
  - In-text citation should be included here
- (20 pts) **Experiment:** A discussion of your hypotheses, methods for trying to answer the question, and how you are conducting those methods.
- (20 pts) **Analysis:** A discussion of the results you gathered and what they mean for your research question.
- (7.5 pts) **Conclusion:** This should act as both a reflection and summary of the project. It should include:
  - A brief, final discussion of the overall results and their impacts
  - A quick reflection on the models and process, including improvements to faults in the project
- (5 pts) **References:** At least three sources must be cited, using APA format.

You will also include R code via one or multiple Quarto/R documents, showcasing the methods you have used.

I will be looking for the following items:

- (10 pts) Appropriate and correct code running through your methods

- Organized?
- Replicable?
- Complete?

The code should follow a logical order with some use of comments/sentences that indicate what processes are being done. No late work will be accepted here.